

CS4545 Assignment 3: Related Work

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RELATED WORK

Nearest neighbor

The Taiwan paper showed that considering both geographic proximity (nearby areas) and functional connections (areas linked by travel and economic activity) improves flu predictions. However, their model focused on measuring accuracy rather than helping public health officials make decisions. My project takes that same idea but applies it to Boston neighborhoods and focuses on answering practical questions:

Which neighborhoods need vaccination clinics next week?
Where should we offer incentives to boost vaccination rates?
This shift from "Can we predict better?" to "How do we use predictions to stop outbreaks?" is crucial because models only matter if they help health departments take action before flu spreads. It also incorporates urban-specific data sources like MBTA ridership and wastewater surveillance to capture functional connections, where as the Taiwan study used county-level correlation patterns[7].

NEIGHBOR'S RELATED WORKS

1st relevant: Graph attention networks

1. This paper introduces Graph Attention Networks (GATs), which use self-attention mechanisms to let nodes assign different importance weights to their neighbors. The method works on both fixed graphs and completely unseen graphs. It achieves state-of-the-art results on citation networks and protein-protein interaction datasets.

2. Novel attention mechanism for graph-structured data that assigns different weights to different neighbors

Multi-head attention architecture that stabilizes learning

Inductive capability—can generalize to completely unseen graphs

State-of-the-art performance on benchmark datasets

3. Application: Citation networks/protein interactions → Public health surveillance (influenza prediction)

Graph type: Static predetermined structures → Dynamic neighborhood networks based on transit and mobility

Task: Node classification → Temporal forecasting (1-2 weeks ahead)

Goal: Benchmark performance → Actionable public health interventions[11].

2nd relevant: Detecting influenza epidemics using search engine query data

1. This paper presents a method for tracking influenza-like illness using Google search query data. The system analyzes 50 million common search queries to estimate ILI physician visit percentages. It provides estimates 1-2 weeks faster

than traditional CDC surveillance with high correlation (0.90-0.97).

2. Automated query selection method that identifies ILI-related searches without prior knowledge

Real-time influenza surveillance system with 1-2 week lead time over CDC reports

Regional and state-level ILI estimates across the United States
Validation showing 0.90-0.97 correlation with CDC surveillance data

3. Data source: Search query text → Structured public health data (wastewater, ED visits, transit ridership)

Spatial resolution: State/regional level → Neighborhood-level predictions

Model architecture: Linear regression on query frequencies → Graph Neural Networks capturing spatial dependencies

Time horizon: Nowcasting (current week estimation) → Forecasting (1-2 weeks ahead prediction)

Graph structure: No explicit spatial modeling → Explicit neighborhood connectivity via transit and demographic ties[2].

3rd relevant: The Parable of Google Flu

1. This paper critiques Google Flu Trends, which overestimated flu prevalence for 100 out of 108 weeks starting in 2011. The authors identify "big data hubris" (treating big data as a substitute for traditional methods) and "algorithm dynamics" (changes to Google's search algorithm affecting predictions) as key failures. They argue that combining GFT with traditional CDC data produces better results than either alone.

2. Documents systematic failure of Google Flu Trends from 2011-2013

Identifies "big data hubris" as assuming big data can replace traditional surveillance

Reveals "algorithm dynamics" problem—platform changes by engineers affect prediction accuracy

Demonstrates that simple lagged CDC models outperform GFT alone

Shows combining GFT with traditional data significantly improves accuracy

Calls for transparency, replicability, and "all data revolution" rather than just big data

3. Lesson application: Critiques overreliance on single data source → Your project integrates multiple diverse data streams (wastewater, ED visits, transit, weather)

Model approach: Identifies overfitting with static correlations → Your GNN explicitly models spatial-temporal dynamics and relationships

Validation strategy: Shows standalone approach failed → Your project combines traditional surveillance with novel data sources

Transparency: Criticizes lack of reproducibility → Academic research with documented methodology

Stability concern: Algorithm changes broke predictions → Your model uses stable public health infrastructure data, not commercial platform data[6].

4th relevant: Economic burden of seasonal influenza in the United States

1. This paper estimates the average annual economic burden of seasonal influenza in the U.S. at \$11.2 billion (\$6.3-\$25.3 billion), with direct medical costs of \$3.2 billion and indirect costs of \$8.0 billion. The study analyzes five outcome categories (non-medically attended illness, office visits, ED visits, hospitalizations, and deaths) across different age groups using 2015 population data. Lost lifetime earnings from deaths represented the largest cost component at 37.2% of the total.

2. Updated economic burden estimate using 2015 demographic and cost data

Comprehensive cost breakdown by age groups and outcome categories

Inclusion of both direct healthcare costs and indirect productivity losses

Estimated 26.2 million annual influenza cases with 247,000 hospitalizations and 36,300 deaths

Sensitivity analysis showing hospitalization rates as most influential variable

Comparison showing their estimates are roughly half of previous 2003 estimates by Molinari et al.

3. Motivation: Quantifies economic burden retrospectively → Your project aims to enable proactive resource allocation

Geographic scope: National/regional estimates → Neighborhood-level predictions in Boston

Time dimension: Annual average burden → 1-2 week ahead forecasting

Application: Cost-effectiveness analysis for policy → Actionable early warning for targeted interventions

Focus: Measuring what happened → Predicting what will happen to prevent it[8].

5th relevant: Graph Neural Networks for Predictive Learning in Urban Computing

1. This survey paper reviews Spatio-Temporal Graph Neural Networks (STGNNs) for predictive learning in urban computing across transportation, environment, public safety, and health domains. The authors systematically examine STGNN architectures including spatial learning networks (spectral/spatial GCNs, GATs), temporal learning networks (RNNs, TCNs, transformers), and fusion methods. They also discuss advanced techniques like adaptive graph learning, meta-learning, self-supervised learning, and physics-informed approaches.

2. Comprehensive taxonomy of STGNNs covering applications, basic architectures, and advanced variants

Categorization of predictive learning tasks across major urban computing domains (60%+ in transportation)

Analysis of graph construction methods (topology, distance, similarity, interaction-based)

Review of spatial learning improvements (multi-graph convolution, adaptive learning, multi-scale)

Examination of temporal learning enhancements (multi-scale, multi-granularity, decomposition)

Discussion of spatio-temporal fusion approaches (factorized, coupled, joint modeling, automated NAS)

Integration with advanced frameworks (adversarial, meta-learning, contrastive learning, physics-informed)

Public dataset compilation and identification of future research challenges

3. Methodological foundation: Broad survey of STGNN techniques → Your project applies specific STGNN architecture to influenza prediction

Application domain: Primarily traffic/transportation focus → Public health (influenza forecasting)

Data integration: Generic spatio-temporal patterns → Multi-modal health surveillance (wastewater, ED visits, transit, weather, vaccination)

Prediction target: Traffic flow, demand, incidents → Disease outbreak intensity 1-2 weeks ahead

Actionability: Retrospective analysis and forecasting → Proactive intervention planning (mobile clinics, targeted campaigns)

Spatial granularity: City-wide or regional → Hyperlocal neighborhood-level predictions for Boston[4].

LOCAL NEIGHBORHOOD RELATED WORKS

6th relevant: Prevention and Control of Seasonal Influenza with Vaccines

1. The paper provides CDC recommendations for preventing seasonal influenza through vaccination in the United States. It reviews influenza epidemiology, vaccine effectiveness, and updates guidance on vaccine composition and use for the 2016–17 season.

2. Establishes national, population-wide influenza vaccination recommendations and eligibility criteria.

Synthesizes surveillance, clinical, and effectiveness data to guide seasonal vaccine policy and public health practice

3. The paper focuses on population-level prevention policy, while your project focuses on neighborhood-level outbreak prediction.

Instead of static recommendations, your work uses graph neural networks and real-time data to generate short-term, localized forecasts that inform targeted interventions before outbreaks peak[3].

7th relevant: The Oregon Child Absenteeism Due to Respiratory Disease Study (ORCHARDS)

1. This paper describes ORCHARDS, a school-based influenza surveillance system combining automated daily absenteeism monitoring in K-12 schools with home visits to ill students for specimen collection. Over 5 years, influenza-like illness caused 6,634 absences (0.20% of school days), and influenza was detected in 23.3% of 1,728 students with acute respiratory infection who received home visits. The study evaluates whether monitoring cause-specific school absenteeism can provide early detection of accelerated influenza transmission in schools and communities.

2. Novel school-based surveillance system integrating automated absenteeism data with home-based specimen collection
 Multi-year longitudinal community study (2014-2019) capturing 3.26 million student-days
 Validation against parallel medically-attended influenza surveillance system in same geographic area
 Household transmission substudy component
 Comprehensive viral testing including multipathogen respiratory panel
 Demonstrated feasibility of continuous cause-specific absenteeism monitoring
 Framework for evaluating school absenteeism as early warning system for influenza outbreaks

3. Data source: School absenteeism reports → Multiple integrated public health data streams (wastewater, ED visits, transit, vaccination)
 Geographic scope: Single school district (Oregon, WI) → Neighborhood-level across Boston
 Surveillance model: School-based with home visits → Population-based without individual specimen collection
 Prediction approach: Descriptive surveillance monitoring → Predictive GNN modeling for 1-2 week forecasting
 Validation method: Parallel clinic-based surveillance → Comparison with retrospective outbreak data
 Primary population: School-aged children (K-12) → All neighborhood residents regardless of age
 Actionable output: Community-wide surveillance signal → Hyperlocal predictions enabling targeted interventions (mobile clinics, outreach)[10].

8th relevant: Data-Centric Epidemic Forecasting

1. This paper is a comprehensive survey of data-centric approaches to epidemic forecasting, emphasizing how diverse data streams and modern machine learning models improve prediction accuracy. It reviews datasets, modeling paradigms (mechanistic, statistical, hybrid), evaluation metrics, and challenges in real-world deployment.

2. Provides a unifying framework for epidemic forecasting that centers data quality, multimodal data integration, and ML/AI methods.
 Systematically categorizes forecasting tasks, datasets, and modeling approaches, highlighting gaps and open challenges in operational public health forecasting.

3. The paper is broad and methodological, while your project is narrow and applied, focusing on neighborhood-level influenza prediction in Boston.
 Instead of surveying models, your work implements a specific spatiotemporal GNN, explicitly encoding transit and socioeconomic networks to produce actionable, hyperlocal forecasts for public health intervention[9].

9th relevant: A contribution to the mathematical theory of epidemics

1. This foundational 1927 paper develops mathematical models for epidemic spread in homogeneous populations, introducing the concept of a threshold population density below which epidemics cannot occur. The authors demon-

strate that epidemics can terminate before exhausting all susceptible individuals, with the epidemic peaking when the unaffected population falls to the threshold density. For diseases with intermediate hosts (like malaria), the product of both population densities must exceed a threshold value.

2. Introduction of threshold density concept—population density below which epidemics cannot sustain
 Mathematical proof that epidemics end before all susceptibles are infected
 SIR-type compartmental model framework (Susceptible-Infected-Removed)
 Derivation showing epidemic size approximately doubles the excess above threshold for small excesses
 Extension to vector-borne diseases with threshold product concept
 Validation using 1905-1906 Bombay plague mortality data
 Foundational differential equations for epidemic modeling

3. Historical context: Foundational theory from 1927 → Modern application with contemporary data and methods
 Model type: Continuous differential equations for homogeneous populations → Graph Neural Networks capturing spatial heterogeneity
 Spatial structure: No spatial variation (homogeneous mixing) → Explicit neighborhood network with transit/demographic connections
 Goal: Theoretical understanding of threshold dynamics → Operational forecasting for public health intervention
 Data requirements: Aggregate population-level parameters → High-resolution multi-source surveillance data
 Assumptions: Homogeneous susceptibility, constant rates → Heterogeneous neighborhoods with time-varying factors (weather, vaccination, mobility)[5].

10th relevant: The role of the airline transportation network in the prediction and predictability of global epidemics

1. This paper develops a stochastic computational framework for global epidemic forecasting using the complete worldwide air travel network (3,100 airports, 17,182 connections) combined with city population data. The authors demonstrate that the heterogeneous structure of airline networks—characterized by heavy-tailed distributions in connectivity and traffic—creates heterogeneous epidemic patterns with reduced predictability compared to homogeneous networks. Using entropy measures and overlap functions, they show that network topology dominates epidemic pathways, with epidemics starting from hub airports showing lower predictability than those from poorly connected cities.

2. Global epidemic model integrating full IATA airline network data with city populations
 Stochastic framework using Langevin formulation for epidemic differential equations
 Quantitative measures for epidemic heterogeneity (entropy) and predictability (overlap function)
 Demonstration that network heterogeneity creates complex epidemic patterns
 Comparison of real network vs. homogeneous and topology-preserving null models
 Finding that connectivity heterogeneity dominates over traffic

heterogeneity in shaping epidemic spread
 Cartogram visualization showing both infection levels and case numbers simultaneously
 3. Network type: Global airline network (3,100 cities world-wide) → Local neighborhood network (Boston only)
 Scale: International pandemic spread → Hyperlocal community transmission within one city
 Transportation mode: Air travel between cities → Transit, walking, demographic mobility within urban area
 Prediction goal: Global pattern heterogeneity and pathways → Neighborhood-level case forecasts 1-2 weeks ahead
 Validation approach: Theoretical analysis with null models → Comparison with actual surveillance data for actionable interventions
 Population mobility: Inter-city air passengers → Intra-city transit ridership and commuting patterns
 Disease compartments: Basic SIR model → Potential for more detailed influenza-specific dynamics with vaccination data[1].

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